

Physicochemical studies of phosphate based P_2O_5 – Na_2O – CaO – TiO_2 glasses for biomedical applications

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Abstract

Glasses P_2O_5 – Na_2O – CaO – TiO_2 with different TiO_2 contents and fixed P_2O_5 (45 wt%) and CaO (24 wt%) have been prepared employing the normal melting and annealing technique. Measurements such as ultrasonic velocity, attenuation, solubility and pH have been carried out in all the compositions of the glasses. It is interesting to note that the above measured ultrasonic parameters exhibit an abnormal behavior (minimum) at 0.5 wt% of TiO_2 content, beyond which an increase in these parameters with increasing TiO_2 content is observed. The maximum pH values and Ca^{2+} ion release have been observed for the TiO_2 free glass those compositions with and the low TiO_2 (≤ 1.0 wt%) content. As the content of the TiO_2 increases, the solubility of the glasses decreases. The observed weight loss reveals two stages of phosphate dissolution kinetics i.e. the first stage, in which the weight loss is proportional to $t^{1/2}$, and a second stage in which a linear behavior is observed.

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1. Introduction

In recent years, applications of bioactive glasses as bioimplants for different biomedical applications have been increasing, particularly in the context of their bone bonding ability with living tissue. The selection of bioactive glasses for implant applications requires the optimisation of their composition [1]. Thus, the preparation of bioactive glasses of different compositions and the characterisation of their structure, stability, solubility and mechanical properties are necessary for the evaluation, and subsequent optimization, of the biocompatibility exhibited by the glass [2,3].

Phosphate based glasses offer unique range of material properties, in view of their tailorable solubility to suit var-

ious hard and soft tissue replacements. They are entirely amorphous if processed correctly. These glasses are made up of an inorganic phosphate network in their main structure, with modifications imposed by the addition of other oxides, such as Na_2O , CaO and TiO_2 [4–6]. Many glasses and glass ceramics used in surgical implants [7] involve the SiO_4^{4-} group as a network forming unit. Silicate based glasses are used as long-term devices to replace hard or even soft tissues due to their insoluble properties. However, the long-term reaction of silica with the tissues is unknown [8]. In addition, phosphate glasses can have controlled rates of dissolution, as compared to silica containing glasses. Due to the inherent chemical and physical properties [9] of phosphate glasses, the potential for application in orthopedic implants is high.

Simple phosphate glasses do not have high chemical stability when compared with multicomponent phosphate based glasses. In addition, soluble phosphate glasses with

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network-modifiers slowly release active ingredients, like calcium ions, during dissolution studies [10]. In this case, such active ingredients are used to cure the damaged tissues when implanted. Therefore, it is necessary to measure the dissolution products and their corresponding weight loss in the as-prepared bioactive glasses in addition to the measurement of pH.

TiO₂ is considered to be harmless in contact with human tissue, and, as a result, has gained interest in the production of bioactive SiO₂–CaO–P₂O₅ glass–ceramics [11–13]. The role of TiO₂ in bioactive glasses and bioactive glass–ceramics is still in debate. TiO₂ is known to induce calcium phosphate surface nucleation in CaO–P₂O₅ based systems [13]. Navaro et al. [14] studied the degradation of TiO₂ stabilised phosphate glasses and showed that the addition of TiO₂ enhanced the chemical durability due to the presence of Ti–O–P bonds rather than P–O–P bonds. In addition, their Raman spectra indicated the presence of TiO₅ and TiO₆ groups.

The determination of elastic properties is also essential as it involves the mechanical interaction between the implant and the surrounding tissues in the newly developed bioactive glasses. However, the elastic properties of bioactive glasses have not been studied systematically in most of bioactive glasses. Such studies would widen the range of options for clinical applications. Ultrasonic non-destructive testing (NDT) is a unique tool for complete characterisation of materials [15–18] such as biological glasses, in addition to the conventional flaw characterisation in industry.

Therefore, in the present investigation, bioactive glasses of P₂O₅–Na₂O–CaO system are prepared containing varying TiO₂ contents with a goal to optimise the composition of the glasses. In order to study the dissolution properties of the prepared bioactive glasses in distilled water, the values of pH and the concentration of calcium ions in solution have been measured systematically. In addition, ultrasonic velocities, attenuation and elastic properties of the different glasses have been studied to reveal the structural changes taking place in the glass system.

2. Experimental procedure

2.1. Preparation of bioactive glasses

Bioactive glasses of the system 45P₂O₅–31Na₂O–24CaO (hereafter, named as PCNT0), 45P₂O₅–30.5Na₂O–24CaO–0.5TiO₂ (hereafter, named as PCNT05), 45P₂O₅–30Na₂O–24CaO–1TiO₂ (hereafter, named as PCNT10), 45P₂O₅–29.5Na₂O–24CaO–1.5TiO₂ (hereafter, named as PCNT15), 45P₂O₅–29Na₂O–24CaO–2TiO₂ (hereafter, named as PCNT20), and 45P₂O₅–28.5Na₂O–24CaO–2.5TiO₂ (hereafter, named as PCNT25) were prepared using normal melting and annealing techniques. The chemicals used in the present investigation were of AR grade with purity 99.99% of Na₂O, CaO, P₂O₅ and TiO₂ and were used without further purification. P₂O₅ was introduced in the form

of ammonium dihydrogen phosphate, while soda and lime were added in the form of their anhydrous carbonates. Titanium was added as pure TiO₂.

The calculated mixed batches were melted in porcelain crucibles. The melting was made in an electric furnace at 1100 °C and the melting was extended from the beginning to 2 h including batch addition until reaching complete to ensure homogeneity. The prepared glasses were cast in pre-heated stainless steel molds and then, the samples were transferred to an annealing muffle furnace regulated at a constant temperature (slightly lower than the T_g value) and the samples were left inside the muffle for almost 2 h. The muffle furnace was then switched off and was left to cool to room temperature at a rate of 25 °C h^{−1}. The same procedure was employed to prepare the remaining composition of glasses. The required size of the glasses for ultrasonic studies and solubility measurements were cut from the above prepared bioactive glasses. The two faces of bioactive glasses were highly polished using automatic polishing machine (Buehler, 60-1996-230, USA). All the bioactive glasses were etched with acetone to remove the foreign particles. The final composition of various concentration of the glasses was checked by atomic absorption spectrometry (Perkin–Elmer, model 372, USA) studies. The analysed chemical compositions were found to be within ± 0.05 mol% of the corresponding theoretical values.

2.2. Density measurements

Archimedes principle was used to measure the density of all glasses employing double distilled water as a buoyant medium. The density of the glasses were obtained employing the relation,

$$\rho = \frac{W_a}{W_a - W_b} \rho_b \quad (1)$$

where W_a is the weight in air, W_b the weight in buoyant and ρ_b the density of buoyant. All the weight measurements were made using a digital balance (Sartorius, USA) with an accuracy of ±0.1 mg. The experiment was repeated five times to get an accurate value in density. The overall accuracy in the density measurement was ±0.5 kg m^{−3}. The percentage of error in the measurement of density was ±0.05%.

2.3. Ultrasonic velocity measurements

The ultrasonic velocity (both longitudinal and shear) measurements were carried out using a high-power ultrasonic pulser receiver system (Fallon Ultrasonic Inc. Ltd., FUI1050, Canada) with a 100 MHz digital storage oscilloscope (HP-54600B) using the cross-correlation technique employing the pulse echo method described from our earlier studies [19]. The longitudinal and shear ultrasonic waves were generated respectively by *X* and *Y*-cut transducers. Precise transit time measurements were carried out by cross-correlation technique described elsewhere

[15]. The precise ultrasonic velocities (U_L and U_S) in each glasses were obtained by measuring the precise transit time (t) using the relation [20]

$$U = \frac{2d}{t} \quad (2)$$

where d is the thickness of the sample. The percentage of error in the measurement of velocity was $\pm 1\%$.

2.4. Attenuation measurements

Attenuation coefficients of the ultrasonic waves (both longitudinal and shear) were measured, at a frequency of 5 MHz using a contact type transducer. A uniform pressure was maintained between the transducer and glass during the contact measurements. Among the various available couplants for room temperature measurements, machine oil and honey were found to be more suitable to get steady back wall echo train in the oscilloscope screen respectively for longitudinal and shear waves. The peak amplitude of successive back wall echoes from the glasses was used to determine the attenuation coefficient using the relation [20],

$$\alpha = \left(\frac{-20}{2(m-n)d} \right) \log \left(\frac{I_m}{I_n} \right) \quad (3)$$

where I_m and I_n are respectively, the maximum amplitude (voltage) of the m th and n th pulse echoes. The percentage error in the attenuation measurement was $\pm 2\%$.

2.5. Solubility testing

The solubility of all glasses were obtained by means of weight loss measurements in deionised water which was incubated at 37 °C. The deionised water of 25 ml was taken into a poly ethylene container. As the solubility is affected by surface area, the surface area of the glass samples was obtained from the measured dimensions using a digital micrometer screw gauge (Mitutoyo digimatic, Japan). The glass samples were then immersed into the container by touching the bottom of the containers. The sample with container was then placed into an incubator at 37 °C. The glass samples were removed from the container at various time points and then, the excess moisture was removed by blotting the samples dry using a tissue paper. The entire measurement was done by keeping the surface/volume ratio of 1/10 as constant as described elsewhere [21]. By knowing the initial weight (M_0) of each sample and the weight loss (M_t) at time t , the weight loss per unit area was obtained as,

$$\text{Weight loss} = \frac{M_0 - M_t}{A} \quad (4)$$

where A is the surface area (cm^2).

The weight loss was measured in all glasses up to 500 h in deionised water. All the measurements were carried out in digital balance (Sartorius, USA). Weight loss per unit

area was plotted against time. The percentage of error in the measurement of weight loss was $\pm 0.001\%$.

2.6. Ion concentration and pH measurements

The pH and Ca^{2+} ion release were measured employing an ion selective electrode (Orion 97-20 ion plus calcium electrode, USA) combined with a pH meter (Thermo Orion, USA) in all glasses. The above measurements were made along with the solubility test, maintaining the same conditions. The calcium ion electrode was calibrated initially using volumetric standard solutions with concentrations of 10, 100 and 1000 ppm. The pH electrode was calibrated using the standards of pH 4.01, 7.0 and 11.0. The measurements were conducted up to 500 h. The percentage of error in the measurement of pH and Ion concentration was $\pm 0.005\%$.

2.7. Mechanical properties

The elastic constants of all glasses were determined [22] from the measured values of density (ρ), longitudinal velocity (U_L) and shear velocity (U_S) using the relations,

$$\text{Longitudinal modulus } L = U_L^2 \rho \quad (5)$$

$$\text{Shear modulus } G = U_S^2 \rho \quad (6)$$

$$\text{Bulk modulus } K = L - \left[\frac{4}{3} \right] G \quad (7)$$

$$\text{Poisson's ratio } \sigma = \frac{(L - 2G)}{2(L - G)} \quad (8)$$

$$\text{Young's modulus } Y = [1 + \sigma] 2G \quad (9)$$

3. Results

The experimental density, velocity and attenuation data along with sample name for the different composition of glasses are given in Table 1. It is inferred from Table 1, that there is an initial decrease in velocity and an increase in attenuation up to 0.5 mol% of TiO_2 , beyond which a continuous increase in velocity and a decrease in attenuation with further addition of TiO_2 is noticed. The composition dependent density shows an initial decrease up to 0.5 mol% of TiO_2 followed by an increase in density with further addition of TiO_2 content as that of velocity as shown in Fig. 1. An initial decrease in longitudinal modulus (L), shear modulus (G), Young's modulus Y and bulk modulus (K) up to 0.5 mol% of TiO_2 beyond which a gradual increase in the same with further addition of TiO_2 are observed as shown in Figs. 2 and 3. On the other hand, the Poisson's ratio (σ) almost remains constant with a small change from 0.278 to 0.275 with increase in TiO_2 content from 0 to 2.5 mol% as observed in Fig. 4.

Fig. 5 represents the plot of weight loss per unit area against time for different TiO_2 contents. It is inferred from Fig. 5 that there is an increase in the percentage of weight

Table 1
 The nominal composition of bioactive glasses with experimental longitudinal (U_L) and Shear (U_s) velocity, longitudinal (α_L) and shear (α_s) attenuation at 5 MHz of P_2O_5 – Na_2O – CaO – TiO_2 at 30 °C

Sample name	Nominal glass composition (wt%)				Velocity (m s ^{−1})		Attenuation (dB cm ^{−1})	
	P ₂ O ₅	CaO	Na ₂ O	TiO ₂	U_L	U_s	α_L	α_s
PCNT0	45	24	31.0	0	4986	2763	0.57	0.17
PCNT05	45	24	30.5	0.5	4982	2760	0.58	0.29
PCNT10	45	24	30.0	1.0	4987	2767	0.56	0.27
PCNT15	45	24	29.5	1.5	4990	2778	0.54	0.25
PCNT20	45	24	29.0	2.0	4993	2781	0.53	0.22
PCNT25	45	24	28.5	2.5	5008	2787	0.51	0.17

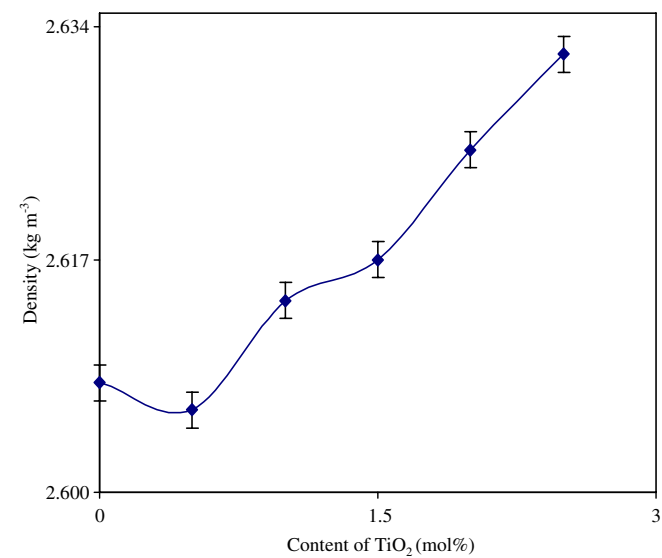


Fig. 1. Dependence of density with change in TiO₂ content on P₂O₅–Na₂O–CaO bioactive glasses (The lines are guides to the eyes).

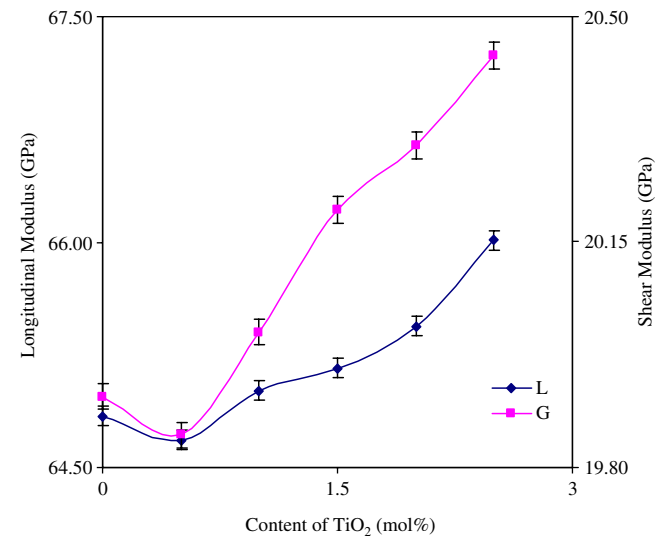


Fig. 2. Dependence of longitudinal (L) and shear modulus (G) with TiO₂ content on P₂O₅–Na₂O–CaO–TiO₂ bioactive glasses (The lines are guides to the eyes).

loss (solubility) with increase in immersion time from 0 to 500 h in all glasses. An initial increase in the leaching of

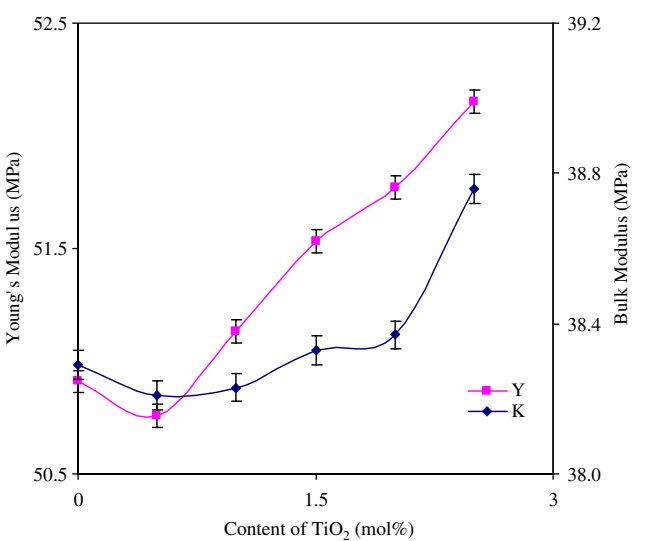


Fig. 3. Dependence of bulk (K) and Young's modulus (Y) with TiO₂ content on P₂O₅–Na₂O–CaO–TiO₂ bioactive glasses (The lines are guides to the eyes).

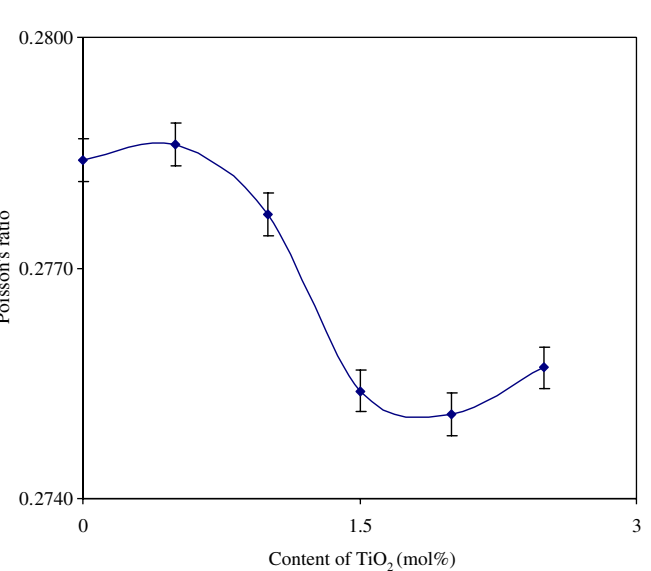


Fig. 4. Variation in Poisson's ratio (σ) with TiO₂ content on P₂O₅–Na₂O–CaO–TiO₂ bioactive glasses (The lines are guides to the eyes).

Ca²⁺ ion (pH value) in glasses PCNT0, PCNT05 and PCNT10 for immersion up to 150 h is observed, followed

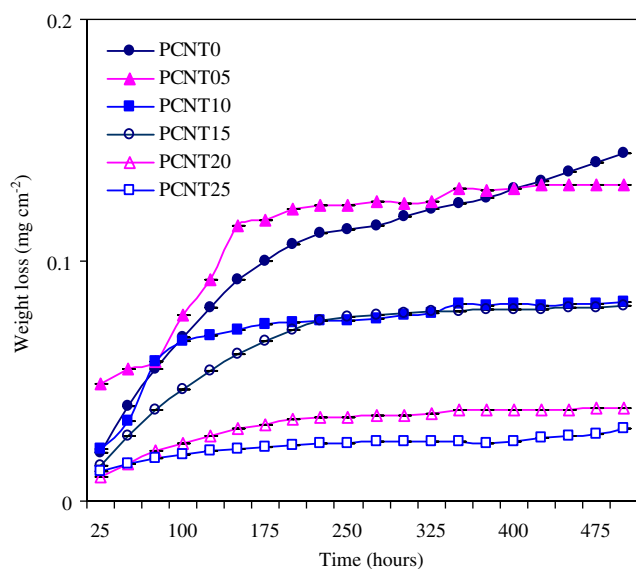


Fig. 5. Weight loss versus dissolution time on P_2O_5 - Na_2O - CaO - TiO_2 bioactive glasses during degradation in deionised water at 37 °C (The lines are guides to the eyes).

by a decrease and some sort of constancy with time as shown in Figs. 6 and 7. On the other hand, a continuous decrease in pH value in PCNT15, PCNT20 and PCNT25 glasses are noticed (Fig. 6).

4. Discussion

In the present glasses, the P_2O_5 and CaO contents remain constant ($P/Ca = 3.0$) while the TiO_2 content varies from 0 to 2.5 mol%. It is well known that density is an

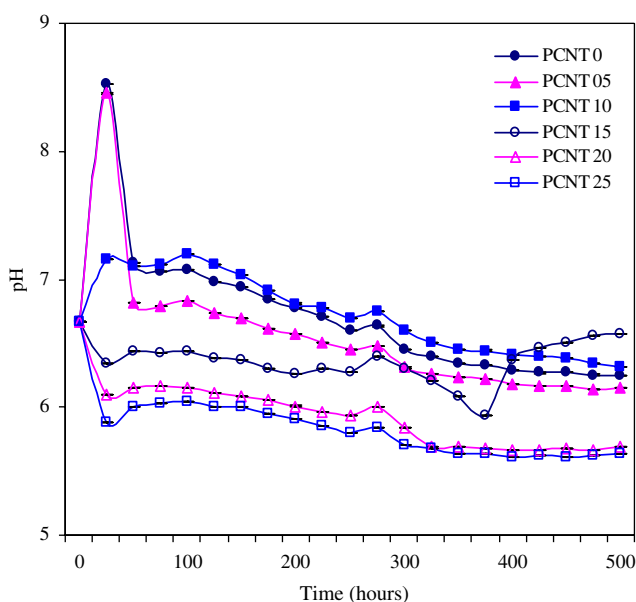


Fig. 6. Effect of TiO_2 content on solution pH on P_2O_5 - Na_2O - CaO - TiO_2 bioactive glasses in distilled water at 37 °C (The lines are guides to the eyes).

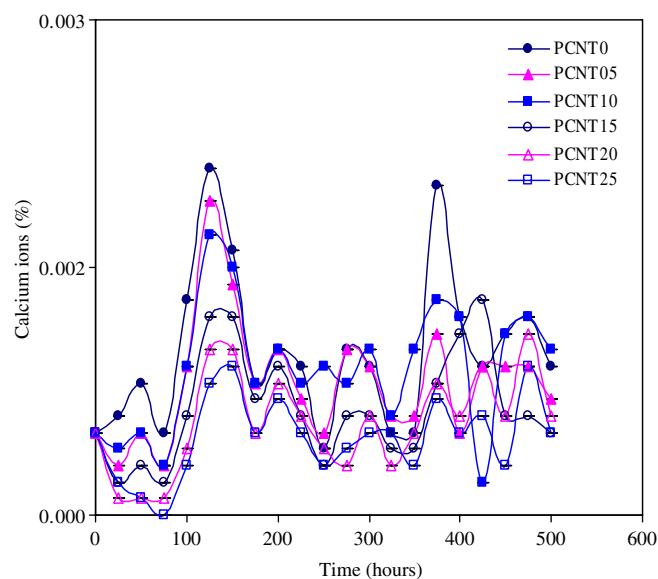


Fig. 7. Effect of TiO_2 content on % of Ca^{2+} ion release on P_2O_5 - Na_2O - CaO - TiO_2 bioactive glasses in distilled water at 37 °C (The lines are guides to the eyes).

important tool [22] to explore the structural compactness/softening, the change in geometrical configurations, coordination number and cross-link density etc. Thus, it helps to reveal the degree of change in the structure with change in composition in any glass systems. Generally, when metal oxides are added into the glass network either a decrease in density or an increase in density will be noticed without any abnormal behavior. However, in the present investigation, the observed anomalous behavior i.e. minimum in density at 0.5 mol% of TiO_2 is interesting. The initial decrease in density with addition of 0.5 mol% of TiO_2 may affect the structure resulting in the loose packing of the atoms. As a result, a decrease in density with addition of TiO_2 (<0.5 mol%) is observed as shown in Fig. 1. A further addition of TiO_2 leads to an increase in density up to 2.5 mol% of TiO_2 content. The increase in density with addition of TiO_2 content leads to a decrease in volume due to the charge polarity of the atoms resulting in compactness of the glass network. Another postulation is that on the first addition of TiO_2 , Ti^{4+} ions seem to participate in a modifying position causing loosening of the network with further increase of TiO_2 beyond 0.5 mol%, the titanium ions are believed to form structural units of TiO_5 (or TiO_4) group and strengthening the network.

The observed initial softening in the glass network up to 0.5 mol% of TiO_2 and compactness of the network beyond 0.5 mol% can be explained in the following ways: An initial introduction of the network modifier namely TiO_2 leads to the breaking of $P-O-P$ bonds phosphate network and formation of terminal oxygens. Thus, the Ti^{4+} ions are connected to PO_4 tetrahedra through non-bridging oxygens (NBO). A study [23] on MIR spectra of phosphate glasses with and without the addition of TiO_2 has been made. The previous results indicate that the existence of

the elongation of P=O and P–O–P bonds with the addition of TiO₂. As a result, an initial decrease in density with addition of TiO₂ (<0.5 mol%) is observed. A further addition of TiO₂ (<0.5 mol%) in the glass network results in the creation of ionic cross-linking between non-bridging oxygens of two different chains, thereby reinforcing the glass structure [23–25]. The size of the introduced ions plays a dominant role in regulating the chemical stability of glasses. Thus, the metallic ions with small ionic radius and high electrical charge contribute to the formation of stronger P–O–M bonds in the P₂O₅–Na₂O–CaO–MO glass system. Several modifying ions such as (M) Fe³⁺, Al³⁺, Zn²⁺ and Ti⁴⁺ have been used for reinforcement in glass system [26]. Among them Ti⁴⁺ has been found to be very effective to improve the stability and mechanical properties of the glasses [27,28]. The observed increase in density of glasses beyond 0.5 mol% of TiO₂ content confirms the strengthening of the glass structure as indicated by the increase in Young's modulus (Figs. 2 and 3).

In the present bioactive glasses, the ratio of P/Ca was kept constant at 3.00. The composition dependence of velocity shows similar observations as that of density and also exhibits a minimum at 0.5 mol% of TiO₂ beyond which a continuous increase in velocity with addition of TiO₂ content is noticed, as shown in Table 1. The longitudinal and shear attenuation show a reverse trend to that of velocity and also exhibit a maximum at 0.5 mol% of TiO₂ (Table 1) content.

The structural changes observed in the present glasses are best described employing the Q^n terminology, where n represents the number of bridging oxygens (BO) for phosphate tetrahedra (PO₄^{3−}) [29]. Where there is no BOs attached to the single PO₄ tetrahedron, it is known as Q^0 single PO₄ tetrahedron or orthophosphate units. Phosphate tetrahedral attached with one BO, it is termed as Q^1 units or pyro phosphate. If the BOs are attached with PO₄^{3−} tetrahedral, then it is called as meta phosphate or Q^2 species. On the other hand, if it consists of three BOs it is known as Q^3 units or ultra phosphate and its structure is a 3D network. Generally, when an alkali oxide is added to phosphate glasses, the structural groups change from Q^3 to Q^2 to Q^1 to Q^0 depending on the change in the ratios of M₂O (or MO)/P₂O₅ = R passes from 0 to 1 to 2 and to 3. The glasses studied of the system P₂O₅–CaO–Na₂O have both Q^2 (PO₂) and Q^1 (PO₃) phosphate groups [29]. The fraction of Q^1 groups increases proportionally to the content of the network-modifiers. In the present glass network, Ti is introduced into the glass mainly in the form of Ti⁴⁺ according to the abundant of modifier oxides. When it is incorporated into the structure as Ti⁴⁺, co-ordinated by four oxygen, it acts as a glass network former oxide (and in recent studies they assume its presence as TiO₅ groups), while it acts as a glass network modifier in octahedral coordination as TiO₆. The observed difference in the anomalous behavior in velocity, modulus and attenuation with change in TiO₂ contents are due to the incorporation of TiO₂ in glass network structure in two co-ordination

namely TiO₄ (or TiO₅) and TiO₆. The initial decrease in velocity, modulus and increase in attenuation is due to the incorporation of titanium as a network modifier (Table 1 and Figs. 2 and 3). In the lower compositions, most of the oxygen present in the form of non-bridging oxygen (NBO) and the glass network is build up of short chains (2,3 tetrahedrons) or rings through the metal cations. This results in the decrease in velocity, modulus and an increase in attenuation. The addition of TiO₂ beyond 0.5 mol% leads to a change in the network from chain extending Q^2 tetrahedra to chain terminating Q^1 units [30] resulting in the enhancement of network structure. In addition, as TiO₂ has small ionic radius and a large charge, it enters into the glass network in a position which create a cross-linking between Ti⁴⁺ ions and the phosphate tetrahedral (Ti–O–P instead of P–O–P). Thus, the enhancement of the structure with addition of TiO₂ beyond 0.5 mol% results in the increase in modulus, velocity and decrease in attenuation as noticed in Figs. 2 and 3, Table 1. It is evident from the recent studies [31,32] that TiO₂ has the ability to enter into the vitreous arrangements and thus, creates a stronger Ti–O–P linkage and increases the O/P ratios in the network are causes depolymerisation in the phosphate network. Thus, the glass network becomes compact and thereby increases elastic moduli as observed in Figs. 2 and 3. The observed change in the elastic moduli explores the change in cross-link density as revealed from other studies [33]. Even though, the Poisson's ratio [20] is an effective tool to explore the change in cross-link density, it remains almost constant in the present glass system. It varies from 0.278 to 0.275 with change in TiO₂ content (Fig. 4).

According to Bunker et al. [32] the dissolution kinetics of the phosphate glass can be divided into two stages namely the first stage in which the weight loss is proportional to $t^{1/2}$, and the second stage in which it shows a linear behavior. In the present investigation, the above two-fold mechanism has been effective with change in composition of TiO₂ and dissolution hours. In the titanium free and low % of titanium containing phosphate glasses, the initial stages of weight loss obey with $t^{1/2}$ relation, i.e. titanium free and 0.5 and 1.0 mol% TiO₂ containing glasses as shown in Fig. 5. However, an increase in TiO₂ content shows a linear relation with dissolution rate (Fig. 5). The composition dependence of weight loss with $t^{1/2}$ in the titanium free and low content of TiO₂ is due to the higher solubility of the glasses than the other glasses having TiO₂ content greater than 1.0 mol%. The observed non-dependence of weight loss with $t^{1/2}$ is due to the less solubility of the glasses when TiO₂ content exceeds 1.0 mol%. The observed results are in accordance with Bunker et al. [32]. On the other hand, it is inferred that an increase in TiO₂ content increases the ionic strength of the leaching solution which in turn decreases the dissolution rate. The decreases in the dissolution rate are ascribed due to the increase in electrostatic interaction which takes place at the hydrated layers due to the increase in ionic strength. The above results confirm that

the dissolution rate of the phosphate glasses decreases with the increase in TiO_2 content.

Considering the % of Ca^{2+} ion release and the pH values of the different composition with varying TiO_2 contents which can further support the previous results. It is interesting to note that the highest level of pH value (Fig. 6) and Ca^{2+} ion release are observed for titanium free and the low percentage of TiO_2 content (≤ 1.0 mol%), where the solubility rate is also high. Fig. 7 reveals that the percentage of Ca^{2+} ions released in the solution is high in titanium free glasses than in 0.5 mol% and 1.0 mol% of titanium containing glasses. The amount of Ca^{2+} ion release decreases as the percentage of TiO_2 increases from 1.0 mol% to 2.5 mol%. However, in PCNT15 at 400 h a larger amount of Ca^{2+} ion release leads to a gradual increase in pH. The observed data of Ca^{2+} ion release with the change in the TiO_2 content are correlated with the solubility rate and as well as with pH values. The observed results are in accordance with the results obtained in ternary based P_2O_5 , CaO and Na_2O glasses [14] with addition of transition metal oxide like TiO_2 leading to the structural rearrangement in the glass network, and subsequent changes in the physical, mechanical and dissolution properties.

A stronger support to the anomalous experimental density and ultrasonic results comes from a recent In-vivo study of the bioactivity by the sharing of one of the authors [34] of the same samples by implanting them in the femurs of a rabbit. This study shows that the addition of 0.5 mol% TiO_2 increases the bioactivity especially on crystallization and that the further increase in the titanium dioxide content decreases the process. Also, it is concluded that the degradation of the bioglass is not the only important factor for the process of bioactivity but the composition of the bioglass also plays an important role for the bone growth on the material.

5. Conclusions

The glasses in the system P_2O_5 – Na_2O – CaO – TiO_2 , were investigated in this study. Compositions with fixed P_2O_5 (45 mol%) and CaO (24 mol%) contents, a constant P/Ca ratio of 3.00 and a TiO_2 content ranging 0–2.5 mol% (replacing Na_2O) were investigated. The observed minimum in density, velocity, modulus and a maximum in attenuation with change in TiO_2 content indicates a softening of glass network up to 0.5 mol% of TiO_2 content beyond which an increase in the compactness of glass network with further addition of TiO_2 content is observed. When a small amount of TiO_2 is added to the glass network it is assumed that TiO_2 can act as a network modifier and hence, leads to the above observed minima at 0.5 mol% of TiO_2 . The further increase in TiO_2 beyond 0.5 mol% leads to change in the network from chain extending Q^2 tetrahedra to chain terminating Q^1 units that contributes to an enhancement in the network structure.

The dissolution rate of the phosphate glasses obeys the two stages of weight loss namely the first stage in which

it is proportional to $t^{1/2}$ and the second stage in which it is linear. The above results confirm that the solubility of the glass is higher in the titanium free and low titania content glasses. The dissolution rate of the phosphate glasses decreases with an increase in TiO_2 content. The measured pH and Ca^{2+} ion release also confirms these results. It is inferred that PCNT05 has a higher bioactivity than higher TiO_2 containing glasses. Moreover, such degradation of the bioglass is not the only important factor contributing to bioactivity.

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